RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description

UNIT I (Satellite Orbits)

Degree, Semester& Branch: VII SEMESTER B.E. ECE B
Course Code & Title: EC6004 Satellite Communication
Name of the Faculty member: Mrs.R.Chitra
Name of the topic: Launching Procedures, Launch vehicles and propulsion
Name of the Activity: Animated Video
Date & Duration: 19.07.2019 & 10 minutes

Description:

Videos are the simplest way to express your complicated ideas in a memorable way, and they are the best tools to enable educational and other kinds of organizations to accomplish this goal. Animated videos can be great in imparting information to people in broadcasting and other public media. Using great story lines and catchy language in animated explainer videos, it's easy to convey important information to all viewers. Animated explainer videos can intelligently, efficiently, and smoothly deliver ideas and information and because they are funny, people remember them for a long time.

Goals (Learning Outcomes):

1. The students will be able to learn the launching procedures of satellites easily.

Use of appropriate methods:

Justification for choosing the topic using Animated Video activity:

The process of placing the satellite in a proper orbit is known as launching process. During this process, from earth stations we can control the operation of satellite. Mainly, there are four stages in launching a satellite. When the satellite reached to the desired height of the orbit, its subsystems like solar panels and communication antennas gets unfurled. Then the satellite takes its position in the orbit with other satellites. Now, the satellite is ready to provide services to the public. Animated Video helps to visualize how the technologies are used to launch satellite into space.

Effective presentation:

Implementation (Plan & Execution) with Proof:

- Animated video for satellite launching procedure was displayed using Projectors and explained about the technologies used in launching satellites.

Sample Images:

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- By conducting this activity, the students were able to understand the concepts and they were able to score good marks for the questions under this topic during Internal Assessment Test I.

Reflective Critique:

Benefits:

- The students are interested to view multimedia clip apart from theory concept to know the visualized concept behind satellite launching.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1					2	1		

CO1: The Students will be able to determine the azimuth and elevation angles and visibility of a geostationary satellite from an earth station

References:

1. https://www.youtube.com/watch?v=VBf_WsHTH_c
2. Text Book: Dennis Roddy, "Satellite Communication", 4th Edition, Mc Graw Hill International, 2006.
3. https://www.isro.gov.in/launchers

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT II (Space Segment and Satellite Link Design)

Degree, Semester & Branch: VII SEMESTER B.E. ECE B
Course Code & Title: EC6004 Satellite Communication
Name of the Faculty member: Mrs.R.Chitra
Name of the Topic: Link Budget
Name of the Innovative Practice: One Minute Paper
Date & Duration: 01.08.2019 & 5 minutes

Description:

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, Students must first organize their thinking to rank the major points and then decide upon a significant question. It is thus a very adaptable tool.

Goals (Learning Outcomes):

1. The students will be able to create link budgets for an uplink and a downlink of satellite communication.
2. The students will be able to analyse satellite link design.

Use of appropriate methods:

Justification for choosing the topic using One Minute Paper activity:

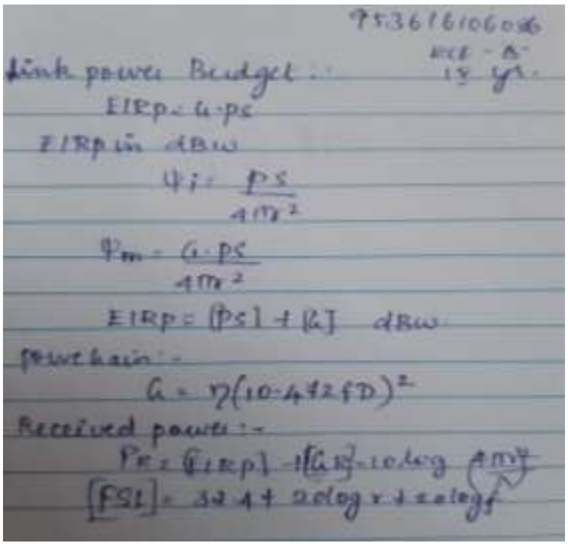
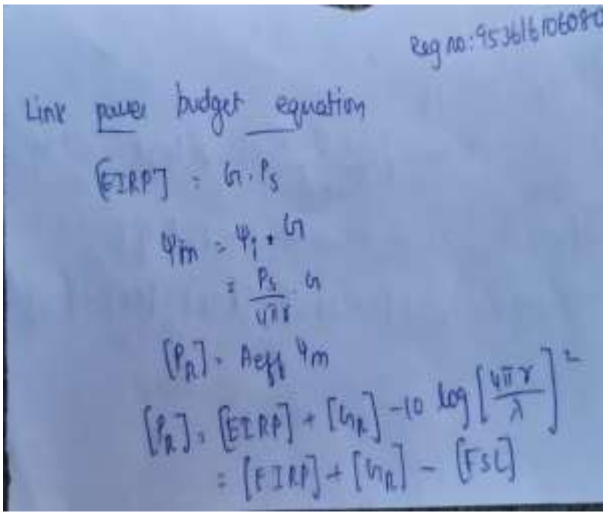
Link budget is an important topic that involves many equations such as carrier to noise ratio, EIRP, Flus density, Back-off etc. As it may be confused, making it as a practice to write those equations is a healthy aspect to remind those equations. As an exam point of view, it is frequently asked topic in University exam.

Effective presentation

Implementation (Plan & Execution) with Proof:

- Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally.
- Each and every student is expected to participate and has something important to contribute - no matter what their cultural background or prior level of academic preparedness

Sample Images:

Link Budget	
	

<u>Significance of Results:</u>

Assessment of Effectiveness/Success of the Activity:

- ➢ The students felt easy to remember and recollect the concepts and equations involved in the topic of link budget which made them to attend the questions under this topic in Internal Assessment Test II and score good marks.
- ➢ They were able to realize the point of small mistakes that they may commit while using the equations.

Reflective Critique:

1. **Benefits:**
   - ➢ Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
   - ➢ I have found that this gives the more reflective chance to the students to gather their thoughts prior to verbalizing them and benefits that student who are more fearful of public speaking by giving them a script to fall back on (or build on)
2. **Challenges:**
   - ➢ As it is mentioned above that it takes one minute to conduct this activity. But, the students are failed to remember all equations in a given duration. They took five minutes to complete this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	1	1				1		2	1	2

CO2: The Students will be able to explain the concept of signal propagation of space segment components and create link budgets for an uplink and a downlink.

References:

1. Text Book: Dennis Roddy, “Satellite Communication”, 4th Edition, Mc Graw Hill International, 2006
2. https://provost.tufts.edu/celt/files/MinutePaper.pdf

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT III (Earth Segment)

Degree, Semester & Branch: VII SEMESTER B.E. ECE B
Course Code & Title: EC6004 Satellite Communication
Name of the Faculty member: Mrs.R.Chitra
Name of the Topic: Transmission Losses
Name of the Innovative Practice: Think-Pair- Share
Date & Duration: 21.08.2019 & 20 minutes

Description:

Think-Pair- Share activity has three parts:

1. Think - The teacher poses a question and gives students thinking time on their own (up to two minutes). Questions with several possible solutions are more likely to generate discussion.

2. Pair - Each pair compares their ideas and reaches a mutually agreed response to the question. Circulate around the class and listen to the conversations.

3. Share - The teacher then gathers ideas across the classroom about student's thinking. Teacher can allow each pair to choose who will present their ideas to the class.

Goals (Learning Outcomes):

1. The students will be able to identify the difference between various types of transmission losses.
2. The students will be able to analyze the effect of transmission losses in a satellite link.

Use of appropriate methods:

Justification for choosing the topic using Think-Pair- Share activity:

In commutations, when the signal is transmitted from one end to other end, losses may occur. The losses which are constant or variable one. No matter what precautions we might have taken, still these losses are bound to occur. Such losses may be inexplicit to understand and to remember. This activity enables them to list the losses and its types without any confusion.

Effective presentation

Implementation (Plan & Execution) with Proof:

- I have planned this activity for 15 minutes
- Initially I have explained what students need to try and asked the students to form a pair
- Made all the students to engage in participation
- Students were discussed about the given topic with each other in talk partners for five minutes.
- After that, they are asked to express their thoughts to an audience and contribute to a shared solution.
- Make sure there is enough time for both students in a pair to discuss their ideas.

Sample Images:

Transmission Losses	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Students were able to remember the content of the topic by conducting this activity and maximum students were attended the questions under this topic in Internal Assessment Test.

Reflective Critique:

Benefits:

- This step ensures that every student has the opportunity to explain their thoughts to others and listen to others conversations.
- Particularly students who are uncomfortable speaking in front of the whole class got chance to express their views also.
- Due to this activity, even though one student leaves any type of losses the other student will help to recollect the ideas. So they felt comfortable to think back to the types of losses and share it.
- Students were able to remember the content of the topic by conducting this activity and maximum students were attended the questions under this topic in Internal assessment Test.

Challenges:

- One of the biggest challenges of the think-pair-share is to get all students to truly be engaged.
- Most pairs shared the same ideas about the topic
- It took more time than what I planned

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	3	2	3				1		2	1	2

CO3: The Students will be able to analyze the effect of rain attenuation in a satellite link and the availability of the link based on geographic location of the earth terminals.

References:

1. Text Book: Dennis Roddy, “Satellite Communication”, 4^{th} Edition, Mc Graw Hill International, 2006.
2. Wilbur L.Pritchard, Hendri G. Suyderhoud, Robert A. Nelson, “Satellite Communication Systems Engineering”, Prentice Hall/Pearson, 2007,
3. www.et.iitb.ac.in › Resources › files › TPS

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT IV (Satellite Access)

Degree, Semester & Branch: VII SEMESTER B.E. ECE B
Course Code & Title: EC6004 Satellite Communication
Name of the Faculty member: Mrs.R.Chitra
Name of the Topic: Revision of Unit IV
Name of the Innovative Practice: Mind Map
Date & Duration: 24.09.2019 & 20 minutes
ICT Tool Used: -

Description:

Mind mapping is a visual form of note taking that offers an overview of a topic and its complex information, allowing students to comprehend, create new ideas and build connections. Through the use of images and words, mind mapping encourages students to begin with a central idea and expand outward to more in-depth sub-topics.

Goals (Learning Outcomes):

1. The students will be able to differentiate the types of multiple access.
2. The students will be able to design satellite communication system using analog or digital modulation technique
3. The students will be able to analyze the various methods of satellite access.

Use of appropriate methods:

Justification for choosing the topic using Mind Map activity:

In unit IV, different types of multiple access methods in Satellite communication are discussed. But, Students were not able to differentiate the types of multiple access method. This mind map activity gives the graphical representation of words, concepts or pictures linked to and arranged around a given topic using a non-linear graphical layout which helps the students to build framework around a central concept.

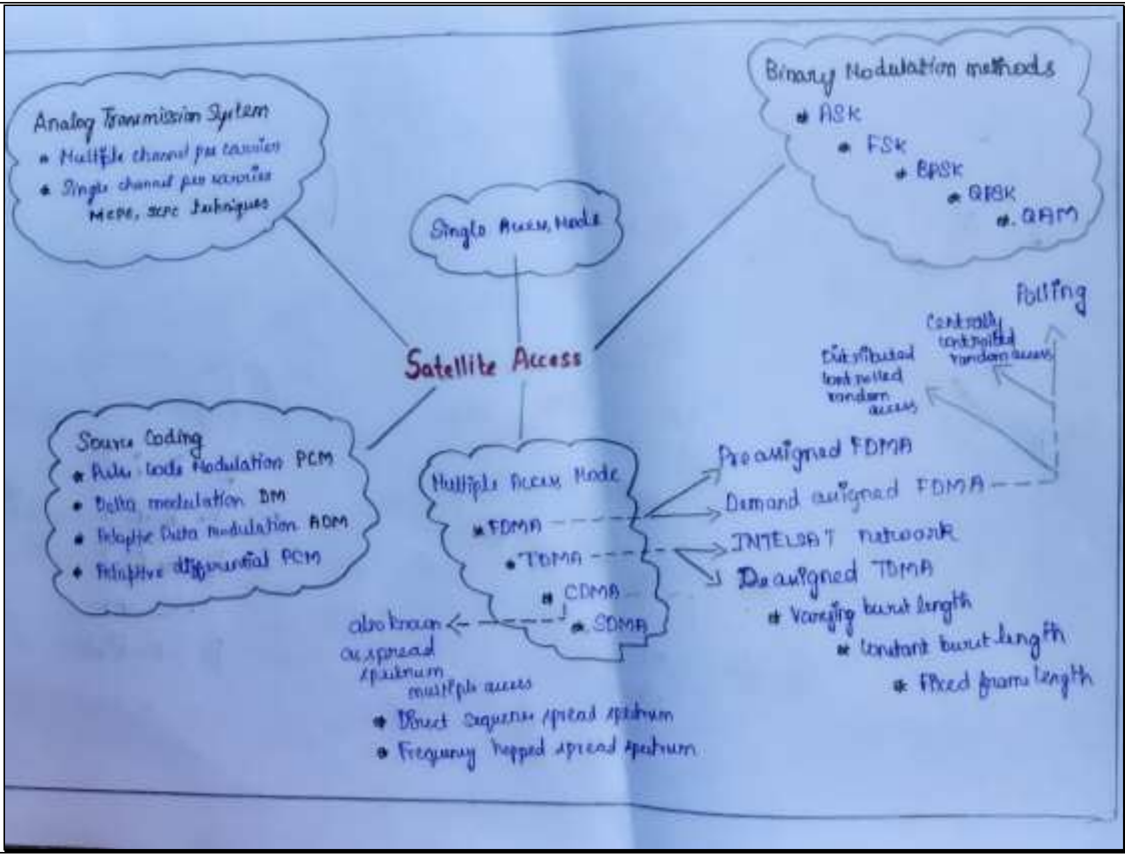
Effective presentation

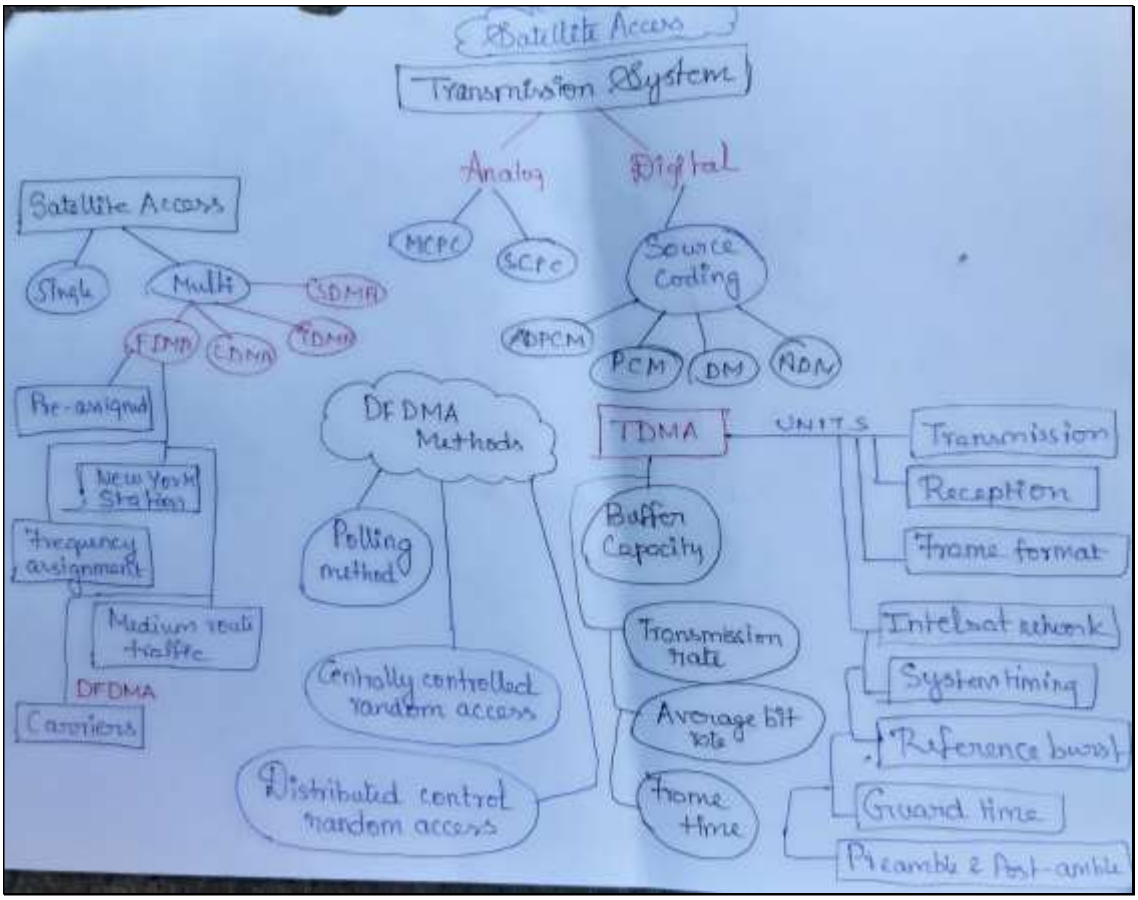
Implementation (Plan & Execution) with Proof:

- I have planned this activity for 30 minutes
- Explained how to do this activity.
- Asked students to sit in a group(4-5 students) and Made all the students to engage in participation
- Generally students know to do this activity for a particular topic. But, I asked them to do this activity for entire unit. So they struggled to start. Later, I have explained how to draw the mind map for the whole unit.
- Students were discussed with each other in group for ten minutes and then started to draw the mind map. Finally, they are asked to express their thoughts to an audience and shared their views.

Sample Images:

Revision of Unit IV





Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ➢ After this activity, students were able to give comparison between FDMA, TDMA and CDMA

Reflective Critique:

Benefits:

- ➢ Help students brainstorm and explore any idea, concept, or problem
- ➢ Facilitate better understanding of relationships and connections between ideas and concepts of satellite multiple access
- ➢ All students were able to recall important concepts and technical terms and take notes

Challenges:

- ➢ Actually I have planned to conduct mind map activity by calling students one by one to come and write their views on board. But, the students hesitated to come in front of the class. So I asked students to sit in group and write their ideas in paper.
- ➢ took more time than what I planned

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO4	2	3	3	2				1		2	1	2

CO4: The Students will be able to design satellite communication system to carry voice, video, data signals using analog or digital modulation

References:

4. http://www.ijsrp.org/research-paper-1215/ijsrp-p4843.pdf
5. Text Book: Dennis Roddy, “Satellite Communication”, 4th Edition, Mc Graw Hill International, 2006.
6. http://www.inspiration.com/visual-learning/mind-mapping

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Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT V (Satellite Applications)

Degree, Semester & Branch: VII SEMESTER B.E. ECE B
Course Code & Title: EC6004 Satellite Communication
Name of the Faculty member: Mrs.R.Chitra
Name of the Topic: Direct Broadcast satellites (DBS)
Name of the Innovative Practice: Multi-media Clip
Date & Duration: 03.10.2019& 10 minutes
ICT Tool Used: LCD Projectors

Description:

Video clips are short clips of video, usually part of a longer recording. The term is also more loosely used to mean any short video less than the length of a traditional television program. While some video clips are taken from established media sources, community or individual produced clips are becoming more common.

Goals (Learning Outcomes):

1. The students will be able to visualize the integrated parts of the satellite broadcast system.
2. The students will be able to apply the concepts of various broadcast methods used in satellite communication.

Use of appropriate methods:

Justification for choosing the topic using Multi-media Clip activity:

Satellite broadcasting is the distribution of multimedia content or broadcast signals usually originate from a station such as a TV or radio station via a satellite and reception by base stations such as small home satellite dishes or by base stations owned by the local cable network for redistribution to their customers. Multimedia clip helps to visualize how the technologies are used to broadcast TV signals from station to customers

Effective presentation:

Implementation (Plan & Execution) with Proof:

- Multimedia clip for Direct Broadcast satellite services was displayed using Projectors and explained about the technologies used in broadcasting services.

Sample Images:

Direct Broadcast satellites (DBS)	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- When asked questions related this topic, 80% of students were attended in Internal Assessment Test III

Reflective Critique:

Benefits:

- The students are interested to view multimedia clip apart from theory concept to know the visualized concept behind satellite broadcast.

Challenges:

- I have planned to play video clip for ten minutes. But, it consumed more time to play relevant videos for better understanding

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1				1		2	1	2

CO5: The Students will be able to analyze the various types of satellite services according to its applications.

References:

1. https://educalingo.com/en/dic-en/film-clip
2. https://www.youtube.com/watch?v=OpkatIqkLO8&t=292s
3. https://www.youtbe.com/watch?v=TgeGYd3ujdo